

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0086**

ASSESSMENT : **COMP0086A7PC**
PATTERN

MODULE NAME : **COMP0086 - Probabilistic and Unsupervised Learning**

LEVEL: : **Postgraduate**

DATE : **13/05/2021**

TIME : **14:30**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2020/21**

Additional material	N/A
Special instructions	N/A
Exam paper word count	N/A

TURN OVER

[intentionally blank]

PROBABILISTIC AND UNSUPERVISED LEARNING
FINAL EXAMINATION
COMP0086 2020–21

This examination contains SIX questions, of which you should answer FIVE.
Each question is worth 20 marks. If you have time, you may attempt all 6 questions, but only your 5 best scores will count towards your final mark. Make sure you manage your time carefully, read through ALL questions first before attempting any, and don't spend too much time on any one question. Calculators are allowed, but are unlikely to be useful. **Always provide justification and show any intermediate work for your answers. A correct but unsupported answer may not receive any marks.**

In the following, bold symbols represent vectors. When asked to provide algorithms, you may provide equations or use MATLAB/Octave notation for matrix algebra. You may assume the availability of standard matrix functions such as `inv`, `pinv`, `eig` or `svd`. You may **not** assume statistical functions, including `mean` and `cov`. If you choose to use MATLAB notation, you will not lose marks for syntactic errors, as long as your intention is clear and correct.

You might find it useful to consult the following table of standard distributions:

Name	Domain	Symbol	Density or Probability fn
Gaussian (Normal)	$\mathbb{R}$	$x \sim \mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$
Multivariate Normal	$\mathbb{R}^D$	$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$	$ 2\pi\Sigma ^{-1/2} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$
Bernoulli	$\{0, 1\}$	$x \sim \text{Bernoulli}(p)$	$p^x (1-p)^{1-x}$
Binomial	$\mathbb{Z}_{0-N}$	$x \sim \text{Binom}(p)$	$\binom{N}{x} p^x (1-p)^{N-x}$
Multinomial	$[\mathbb{Z}_{0-N}]^D$	$\mathbf{x} \sim \text{Multinom}(\mathbf{p})$	$\frac{N!}{x_1! x_2! \dots x_D!} \prod_{d=1}^D p_d^{x_d}$
Poisson	$\mathbb{Z}_{0+}$	$x \sim \text{Poisson}(\mu)$	$\mu^x e^{-\mu} / x!$
Beta	$[0, 1]$	$x \sim \text{Beta}(\alpha, \beta)$	$\frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$
Exponential	$\mathbb{R}_+$	$x \sim \text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$
Gamma	$\mathbb{R}_+$	$x \sim \text{Gamma}(\alpha, \beta)$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$
Student-t	$\mathbb{R}$	$x \sim \text{Student-t}(\alpha)$	$\frac{\Gamma(\frac{\alpha+1}{2})}{\sqrt{\pi\alpha}\Gamma(\frac{\alpha}{2})} \left(1 + \frac{x^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}$
Inverse-Wishart	$\mathbb{S}_+^p$	$X \sim \text{InvWish}(\Psi, \nu)$	$\frac{ \Psi ^{\frac{\nu}{2}}}{2^{\frac{\nu p}{2}} \Gamma_p(\frac{\nu}{2})} X ^{-\frac{\nu+p+1}{2}} e^{-\frac{1}{2} \text{Tr}[\Psi X^{-1}]}$
Dirichlet	$[0, 1]^D$	$\mathbf{x} \sim \text{Dirichlet}(\boldsymbol{\alpha})$	$\frac{\Gamma(\sum_{d=1}^D \alpha_d)}{\prod_{d=1}^D \Gamma(\alpha_d)} \prod_{d=1}^D x_d^{\alpha_d-1}$

1. A Modified EM for Mixtures

Consider a generic latent variable model, with parameters θ and latent variables $\mathcal{Z}$, which seeks to model observations $\mathcal{X}$. Recall that the free energy of the model is

$$\mathcal{F}(\theta, q(\mathcal{Z})) = \langle \log P(\mathcal{X}, \mathcal{Z}|\theta) \rangle_q + \mathbf{H}[q(\mathcal{Z})].$$

- (a) Prove that $\mathcal{F}(\theta, q)$ is never greater than the log-likelihood of the parameters θ .
For what value of q is the inequality tight? *[4 marks]*

Consider the introduction of a new parameter $\beta \in [0, 1]$ to the free energy:

$$\mathcal{F}_\beta(\theta, q(\mathcal{Z})) = \beta \langle \log P(\mathcal{X}, \mathcal{Z}|\theta) \rangle_q + \mathbf{H}[q(\mathcal{Z})].$$

Suppose, by analogy to EM, we were to maximise $\mathcal{F}_\beta$ alternately with respect to θ and q .

- (b) Derive the general forms of the corresponding E- and M-steps. Highlight the differences and similarities to the standard algorithm. *[8 marks]*

Now consider a mixture of two Gaussians in which each component has a fixed, unit (i.e. identity) covariance matrix. Thus the only parameters of the model are the two means $\{\mu_1, \mu_2\}$ and the mixing proportions $\{\pi_1, \pi_2\}$.

- (c) Derive the β -modified responsibilities computed under the E-step for this model. *[4 marks]*
- (d) Show that after this E-step, the free energy is equal to

$$F_\beta(\theta, q^*) = \sum_i \log \sum_m \pi_m^\beta e^{-\frac{1}{2}\beta \|x_i - \mu_m\|^2} + \text{constant},$$

a form which is analogous to the mixture likelihood. *[4 marks]*

2. Sampling algorithms

Consider a real-valued function f and a probability density p . For a given proposal distribution q , denote by F_m the importance sampling estimate computed using m independent draws from q . We had discussed in class that q yields good results for arbitrary functions f if it is “similar” to p . Suppose we are only interested in estimating $\mathbb{E}_p[f(X)]$ for one specific f . Then we can ask for a q that is well-suited for this particular f . A common criterion for “well-suited” is to ask that the variance of estimates is small.

- (a) Show that the variance of F_m is

$$\text{Var}_q[F_m] = \mathbb{E}_q[f(X)^2] - \mathbb{E}_p[f(X)]^2.$$

[3 marks]

- (b) Show that the variance satisfies the lower bound

$$\text{Var}_q[F_m] \geq \mathbb{E}_p[|f(X)|]^2 - \mathbb{E}_p[f(X)]^2.$$

[4 marks]

- (c) We want to choose a proposal distribution for which the variance in (a) is small, using the bound in (b). For which proposal density q_{opt} is the lower bound attained?

[5 marks]

A popular approach to modeling uses so-called truncated Gaussian distributions, which are Gaussian densities that are restricted to some subset of their domain, and renormalized. The main drawback to this approach is that most computations that would have closed-form solutions for untruncated Gaussians now require numerical integration. Suppose we truncate a Gaussian density g , with mean 0 and variance σ^2 , by restricting it to the interval $[0, 1]$. We denote the restriction of the function g to the interval $\tilde{p}$, and write $Z_p := \int_0^1 \tilde{p}(x)dx$.

- (d) Suppose the constant Z_p is given, and we rejection-sample $\tilde{p}$ using a uniform proposal density q on $[0, 1]$. Let N be the (random) number of draws made from q until we have collected 1000 accepted samples. What is the optimal value of $\mathbb{E}[N]$ we can achieve, as a function of Z_p ?
- (e) Now suppose G is a Gaussian in D dimensions, and we truncate it by restricting to the unit ball $B = \{\mathbf{x} \in \mathbb{R}^D \mid \|\mathbf{x}\| \leq 1\}$, with normalization constant Z_p . Let f be a real-valued function defined on B . If D is large, obtaining Z_p by numerical integration can be hard. Specify an estimator for $\mathbb{E}_p[f(X)]$ that does not require Z_p as input, based on n i.i.d. draws from the uniform distribution q on $[-1, 1]^D$.

[3 marks]

3. Optimisation

- (a) A *perceptron* in $\mathbb{R}^D$ is a two-class classifier $f_{\theta,c} : \mathbb{R}^D \rightarrow \{-1, +1\}$ with parameters $\theta \in \mathbb{R}^D$ and $c \in \mathbb{R}$, given by $f_{\theta,c}(\mathbf{x}) = \text{sign}(\langle \mathbf{x}, \theta \rangle - c)$. The parameters are determined from data $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^D$ with class labels $y_1, \dots, y_n \in \{-1, +1\}$ by minimising the cost function

$$C_n(\theta) := \frac{1}{n} \sum_{i \leq n} \mathbb{I}\{f(\mathbf{x}_i) \neq y_i\} |\langle \mathbf{x}_i, \theta \rangle - c|.$$

If we would drop the last term, we would obtain $\hat{R}_n(\theta) := \frac{1}{n} \sum_{i \leq n} \mathbb{I}\{f(\mathbf{x}_i) \neq y_i\}$, which is just the empirical risk for 0—1 loss, and hence a much more natural criterion than C_n . Why is C_n a better choice than $\hat{R}_n$?

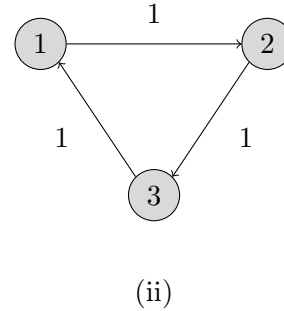
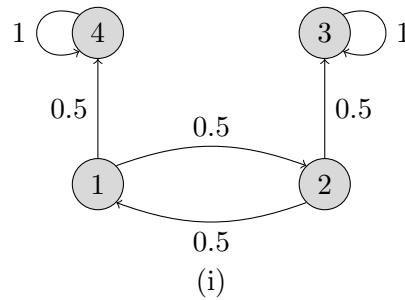
Note: This problem requires no knowledge of classification. [3 marks]

- (b) The function $f(x, y) = xy$ constrained to the unit circle $T = \{(x, y) | x^2 + y^2 = 1\}$ has two maxima and two minima. Determine the locations of the maxima and the locations of the minima. [6 marks]
- (c) Given is some $a > 0$. Suppose we have a routine that evaluates the exponential function numerically. Specify a function f such that gradient descent applied to f terminates with the numerical value of $\ln(a)$. Show that (if we neglect numerical errors) gradient descent on f is guaranteed to terminate with the value $\ln(a)$, for any initial value in $(0, \infty)$. [5 marks]
- (d) Let X be a random variable in $\mathbb{R}^D$, with strictly positive density p . Show that the so-called information inequality $\mathbb{E}[\log q(X)] \leq \mathbb{E}[\log p(X)]$ holds for every strictly positive density q on $\mathbb{R}^D$. [2 marks]
- (e) Consider the multinomial distribution P with parameter vector $p = (p_1, \dots, p_D)$ (see page 2). Determine the maximum likelihood estimator $\hat{p} = (\hat{p}_1, \dots, \hat{p}_D)$.

Note: The multinomial is a distribution on D categories. The data $x_1, \dots, x_D$ are non-negative integers. Each value x_d specifies that category d has been observed x_d times. The total number of observations is $x_1 + \dots + x_D = N$. The parameter is a distribution on D outcomes, i.e. a vector with $p_d \geq 0$ and $p_1 + \dots + p_D = 1$. [4 marks]

4. Markov chains and MCMC

- (a) Does either of the following two Markov chains have an invariant distribution? If so, is it unique?



[3+3 marks]

- (b) Recall that the PageRank algorithm uses an interpolation parameter α to compute an $d \times d$ -transition matrix of the form

$$(1 - \alpha)T + \frac{\alpha}{d} \begin{pmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{pmatrix}.$$

What happens to the PageRank result (the scores assigned to web pages) if we set $\alpha = 1$? [2 marks]

- (c) Suppose we are given two coins A and B , which we regard as random variables $A, B \in \{0, 1\}$. The distributions of A and B are given by the scalars

$$p_A := \Pr\{A = 1\} \in [0, 1] \quad \text{and} \quad p_B := \Pr\{B = 1\} \in [0, 1].$$

We produce a random sequence $X_1, X_2, \dots$ as follows:

- We begin by flipping coin A .
- At each step i , we select the same coin as flipped in the previous step $i - 1$ with probability 0.8, or change to the other coin with probability 0.2.
- X_i is generated by flipping the selected coin.

Define a hidden Markov model to model the sequence $X_1, X_2, \dots$. Please make sure that you specify:

- The state space, and the interpretation of each state.
- The random variables which make up the model.
- All parameters required to determine the model.

[4 marks]

-
- (d) Are the variables X_i and X_{i+2} in the previous question (for any $i \in \mathbb{N}$) stochastically dependent? *[2 marks]*

Consider a Metropolis-Hastings algorithm for a D -dimensional Gaussian target distribution p , with mean 0 and covariance matrix $\sigma^2 \mathbf{I}$, where $\mathbf{I}$ is the identity matrix. The proposal density $q(\mathbf{x}_{i+1}|\mathbf{x}_i)$ is Gaussian with mean $\mathbf{x}_i$ and covariance matrix $\tau^2 \mathbf{I}$.

- (e) To keep notation simple, write $\mathbf{y} = \mathbf{x}_i$ and $\mathbf{z} = \mathbf{x}_{i+1}$. Show that

$$\mathbb{E}_{q(\bullet|\mathbf{y})} \left[\log \frac{p(\mathbf{z})}{p(\mathbf{y})} \right] = -D \frac{\tau^2}{2\sigma^2}$$

[3 marks]

- (f) Explain what the result in (e) implies for the efficiency of the sampling algorithm in two cases: If $D = 2$ (where you should consider the cases $\tau \gg \sigma$ and $\sigma \gg \tau$), and if D is large (that is: much larger than the ratio of τ and σ). *[3 marks]*

5. Bradley-Terry Choice Model

Choice models are statistical models of people's choice behaviour, and are important in econometrics, psychology etc. The Bradley-Terry model is a popular example. In this question, we will derive inference algorithms for this model.

Suppose we have N items (say movies), indexed by $i = 1, \dots, N$. We model the desirability of item i using a single number $\theta_i > 0$. Suppose we ask a person to choose between two items i and j . We can model his choice $x \in \{i, j\}$ using a generative model which simply says that each item has probability of being chosen proportional to its desirability.

Suppose our data consists of M such paired choices, with $x_m \in \{i_m, j_m\}$ being the preferred item among i_m and j_m for $m = 1, \dots, M$.

- (a) Write down the probability of the observed choices $x_1, \dots, x_M$ conditioned on the desirabilities. *[3 marks]*
- (b) What is a “likelihood”? If the previous question had been “Write down the *likelihood* of the observed choices...”, what would have been wrong? *[3 marks]*

To learn the desirabilities from the choice data, we can introduce auxiliary variables. For each pair $\{i, j\}$ let n_{ij} be the number of comparisons between i and j , i.e. with $\{i_m, j_m\} = \{i, j\}$. Let n_i be the number of times item i was chosen in the data. Define auxiliary variables

$$A_{ij} \sim \text{Gamma}(n_{ij}, \theta_i + \theta_j),$$

each independent of the data and other auxiliary variables.

- (c) Show that the log joint probability of the auxiliary variables and data is

$$\mathcal{L} = \left(\sum_{i=1}^N n_i \log \theta_i - \sum_{j \neq i} A_{ij} \theta_i \right) + \left(\sum_{i < j} (n_{ij} - 1) \log A_{ij} - \log \Gamma(n_{ij}) \right)$$
[5 marks]

- (d) Give a prior for the desirabilities that is conjugate to the joint probability of latent variables and data. *[3 marks]*
- (e) Derive a Gibbs sampler to perform Bayesian inference in the resulting model. *[6 marks]*

6. Change Point Detection

Consider a sequence $x_1, \dots, x_T$. Change point detection is the task of determining a point t in the sequence such that $x_1, \dots, x_t$ has a significantly different distribution to $x_{t+1}, \dots, x_T$.

In this question we will assume that $x_1, \dots, x_t$ are iid and can be modelled using an exponential family (and the same for $x_{t+1}, \dots, x_T$). We also assume that the location of the change point is uniformly distributed between 1 and $T - 1$.

- (a) Define an exponential family and a conjugate prior. *[2 marks]*
- (b) Assuming that we use an exponential family with a conjugate prior, and a uniform prior over the change point, describe an algorithm that will compute the posterior distribution over the location of the change point in time linear in T . *[6 marks]*

Suppose now we have a latent variable model, so that each x_i has an associated latent variable z_i . Assume that we have an exponential family model over each (x_i, z_i) with conjugate prior, and that conditioned on the change point and parameters, (x_i, z_i) for different i 's are independent.

- (c) Derive a MCMC algorithm to sample the location of the change point. The latent variables involved are $z_1, \dots, z_T$, the change point location C , and the parameters θ_1 for $x_1, \dots, x_t$ and θ_2 for $x_{t+1}, \dots, x_T$.
Use a Metropolis-Hastings proposal that moves the change point by one step left or right, updating the latent variable of the single data item that is affected by this move. Derive the proposal distribution and acceptance probability. *[8 marks]*
- (d) Discuss how the other variables can be updated. *[4 marks]*